

$$PQ = \begin{pmatrix} 1 & 2 & \dots & n \\ p_1 & p_2 & \dots & p_n \end{pmatrix} \begin{pmatrix} 1 & 2 & \dots & n \\ q_1 & q_2 & \dots & q_n \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & \dots & n \\ p_{q_1} & p_{q_2} & \dots & p_{q_n} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ p \end{pmatrix} \begin{pmatrix} 1 & 3 \\ q \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 & 2 \end{pmatrix}$$

Multiplication table of S_3

S_3	e	(132)	(123)	(23)	(13)	(12)
e	e	(132)	(123)	(23)	(13)	(12)
(132)	(132)	(123)	e	(12)	(13)	(23)
(123)	(123)	e	(132)	(13)	(23)	(12)
(23)	(23)	(12)	(13)	e	(123)	(132)
(13)	(13)	(23)	(12)	(132)	e	(123)
(12)	(12)	(13)	(23)	(123)	(132)	e

$$(23)(132) \rightarrow \begin{matrix} 1 \rightarrow 3 \rightarrow 2 \\ 2 \rightarrow 1 \rightarrow 1 \\ 3 \rightarrow 2 \rightarrow 3 \end{matrix}$$

The Dihedral group

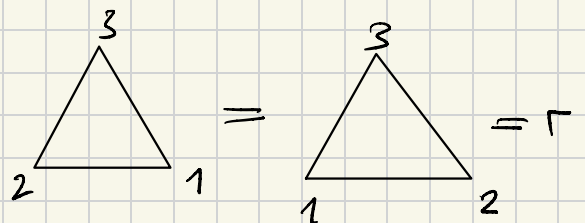
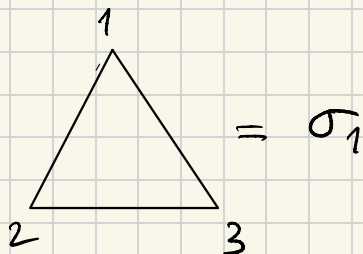
$$D_n \longrightarrow \begin{cases} \bullet \text{ Rotations} & \text{angle } \frac{2\pi}{n} \\ \bullet \text{ Reflection} \end{cases}$$

Presentation $\hookrightarrow |D_n| = 2n$

$$D_n = \langle r, \sigma \mid r^n = e, \sigma^2 = e, \sigma r \sigma = r^{-1} \rangle$$

$$D_3 = \{ \underbrace{e, r, r^{-1}}_{\text{rotations}}, \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{reflections}} \}$$

$$(\sigma_1 \sigma_2)$$



Multiplication table of D_3

D_3	e	r	r^{-1}	σ_1	σ_2	σ_3
e	e	r	r^{-1}	σ_1	σ_2	σ_3
r	r	r^{-1}	e	σ_3	σ_1	σ_2
r^{-1}	r^{-1}	e	r	σ_2	σ_3	σ_1
σ_1	σ_1	σ_2	σ_3	e	r	r^{-1}
σ_2	σ_2	σ_3	σ_1	r^{-1}	e	r
σ_3	σ_3	σ_1	σ_2	r	r^{-1}	e

Generators and rank of a group

Def. of Generator

A subset $S \subseteq G$ is called a set of generators of G if every element of G can be written as a finite product of elements of S and their inverses

$$G = \langle S \rangle$$

Ex / Cyclic group C_3

$$C_3 = \{ e, g, g^2 \} \quad g^3 = e$$

$$\langle g \rangle = C_3$$

$$\text{rank}(C_1) = 1$$

Ex Dihedral $D_3 = \{e, \Gamma, \Gamma^{-1}, \sigma_1, \sigma_2, \sigma_3\}$

$$\sigma = \sigma_1$$

$$\sigma_2 = \Gamma \sigma_1 = \Gamma \sigma$$

$$\sigma_3 = \Gamma^2 \sigma = \Gamma^{-1} \sigma$$

$$D_3 = \langle \Gamma, \sigma \rangle$$

$$\text{rank}(D_3) = 2$$

Group morphism

Homomorphism

G (comp. law $\circ$)

H (comp. law $\diamond$)

A homomorphism is a mapping

$$f: G \rightarrow H$$

$$g \rightarrow h = f(g)$$

Such that $g_1 \circ g_2 = g_3 \in G$

then $f(g_1) \diamond f(g_2) = f(g_3) \in H$

$$D_3 = \{e, \Gamma, \Gamma^{-1}, \sigma_1, \sigma_2, \sigma_3\}$$

$$C_2 = \{e, c\}, \quad c^2 = e$$

$$f: D_3 \rightarrow C_2$$

$$e, \Gamma, \Gamma^{-1} \rightarrow e$$

$$\sigma_1, \sigma_2, \sigma_3 \rightarrow c$$

$$\textcircled{\star} \sigma_1 \circ \sigma_2 = \Gamma$$

$$\Rightarrow f(\sigma_1 \circ \sigma_2) = f(\Gamma) = e(c_2)$$

$$f(\sigma_1) \diamond f(\sigma_2) = c \diamond c = e(c_2)$$

✓

$$\textcircled{\star} \quad \Gamma \circ \sigma_1 = \sigma_2$$

$$f(\Gamma \circ \sigma_1) = f(\sigma_2) = e$$

$$f(\Gamma) \diamond f(\sigma_1) = e \diamond e = e \quad \checkmark$$

Counter example

$$\begin{cases} e & \sigma_3 & \Gamma^{-1} \rightarrow e \\ \sigma_1 & \sigma_2 & \Gamma \rightarrow e \end{cases}$$

$$\sigma_3 \circ \Gamma^{-1} = \sigma_2$$

$$f(\sigma_3 \circ \Gamma^{-1}) = f(\sigma_2) = e$$

$$f(\sigma_3) \diamond f(\Gamma^{-1}) = e \diamond e = e \quad \times$$

Kernel

Kernel of a homomorphism f measures "how much" of the structure of G collapses on the identity of the map f .

$$\ker(D_3) = \{e, \Gamma, \Gamma^{-1}\}$$

Isomorphism

$\hookrightarrow$ bijective homomorphism !

$$G \cong H$$

For finite group isomorphisms $\equiv$ identical multiplication table

$$C_4 = \{e, a, a^2, a^3\}, \quad a^4 = e$$

$$G_1 = \{1, i, -1, -i\}$$

$$\Rightarrow C_4 \cong G_1 \cong G_3$$

$$G_3 = \{1, i\sigma_2, -1, -i\sigma_2\}, \quad \sigma_2 = \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix}$$

Endomorphism

↳ homomorphism of a group G on itself $f: G \rightarrow G$

Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x$
(with addition)

Automorphism

↳ bijective endomorphism

The set of all automorphism of a group G is denoted

$$\text{Aut}(G) = \{ f: G \rightarrow G \mid f \text{ bijective and } f(gh) = f(g)f(h) \}$$

Ex/ group $(\mathbb{Z}, +) \rightarrow$ two automorphism $f(n) = n$
 $f(n) = -n$

Therefore $\text{Aut}(\mathbb{Z}, +) \simeq C_2$

Cayley's Theorem

Every finite group of order n is isomorphic to a subgroup of the symmetric group S_n